TED ( 10 ) 5025 Reg No

Signature

MODEL QUESTION PAPER

SIXTH SEMESTER DIPLOMA ENGINEERING / TECHNOLOGY

**Computer Aided Design & Computer Aided Manufacturing**

MAXIMUM MARKS 100 TIME 3 HOURS

**PART A**

I Answer all questions in one or two sentences. TWO marks each.

1. List out the types of monitors used in practice.
2. List out the THREE sub units of CPU.
3. Define computer aided process planning.
4. Define DNC Machine.
5. Define subroutine.

**PART B**

II Answer any FIVE questions from the following. SIX Marks each.

1. Explain ( DVST ) direct view storage tube.
2. List out the advantages of CAD
3. Describe product cycle ?
4. What is reference point ? Explain.
5. Explain master production schedule
6. Explain (APT) Automatically programmed tolls ?
7. Explain N C word in part program.

## PART C

Answer one full question from each module. 15 marks each.

### Module I

III ( a ) Explain the activity of CAD in manufacturing. 7

( b ) Explain the different types of printers used in practice. 8

**OR**

IV (a) Explain Different types of geometric models. 7

(b) Explain the structure of C P U with block diagram. 8

### Module II

V (a) Explain the different types of Automation. 7

(b) List out the areas where the NC technology can be used. 8

**OR**

VI (a) Explain Aggregate production process. 6

(b) List out the advantages of computer aided process planning (CAPP). 9

### Module III

VII (a) Classify NC machines based on the type of motion. 6

(b) Sketch and Explain the universal type CNC machine. 9

**OR**

VIII (a) Explain the types of machine centers used in NC machines. 6

(b) Explain machine axis conventions. 9

## Module IV

IX (a) What is sequencing. List out the different steps involved in step turning. 5

(b) Write any TEN preparatory functions and their G codes. 10

**OR**

X (a) Write any FIVE NC words used in a part program. 5

(b) Write a NC program of a step turning for the given drawing. 10